Questions

Q1.

A student lifts a toy car from a bench and places the toy car at the top of a slope as shown in Figure 16.

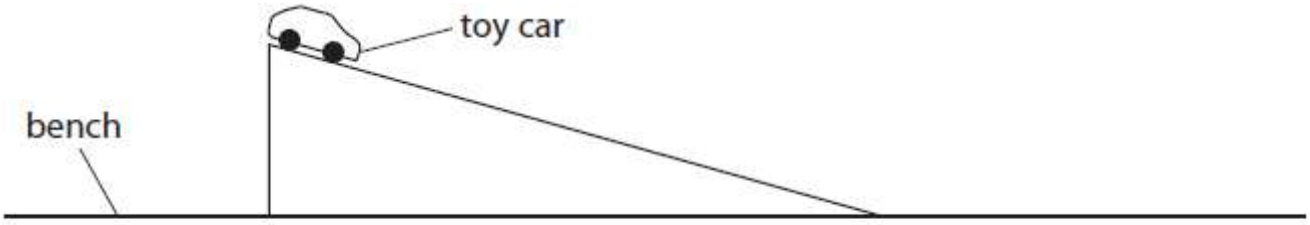

Figure 16

Describe an energy transfer that occurs when the student lifts the toy car from the bench and places the toy car at the top of the slope.

(2)

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(Total for question = 2 marks)

Q2.

As the object in Figure 8 falls, energy is transferred.

These energy transfers are represented in Figure 9.

Complete the labels of the energy stores involved.

(3)

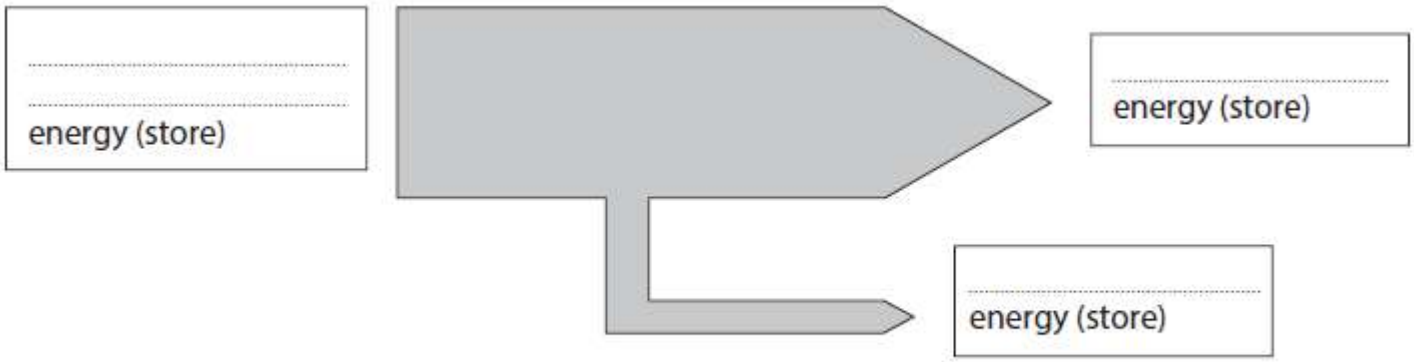

Figure 9

(Total for question = 3 marks)

Q3.

Some questions must be answered with a cross in a box(☒). If you change your mind about an answer, put a line through the box(☒)and then mark your new answer with a cross(☒).

Which statement describes conservation of energy in a closed system?

(1)

☐ **A** when there are energy transfers, the total energy reduces
☐ **B** when there are energy transfers, the total energy does not change
☐ **C** when there are no energy transfers, the total energy reduces
☐ **D** when there are no energy transfers, the total energy increases

(Total for question = 1 mark)

Q4.

(i) The car is moving at 90 km/h when the driver has to stop.

Calculate the thinking time of the driver.
Using the equation:
time = distance ÷ average speed

(2)

thinking time = ... s

(ii) A car has a mass of 1300 kg.

Calculate the kinetic energy of the car when it is travelling at 20 m/s.

(2)

kinetic energy = ... J

(Total for question = 4 marks)

Q5.

Shot-put is an Olympic event.
The shot is a heavy ball.
An athlete throws the shot as far as possible.

A sports scientist analyses an athlete's throw to help improve performance.

In one throw, the shot continues to rise by another 1.3 m after it leaves the athlete's hand.
The mass of the shot is 7.26 kg.

(i) Calculate the amount of gravitational potential energy gained by the shot.

(2)

gravitational potential energy gained = ... J

(ii) Explain how the total energy stored in the shot changes between leaving the athlete's hand and hitting the ground.

(2)

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(Total for question = 4 marks)

Q6.

Figure 18 shows a person on a skateboard at the top of a ramp.

At P, the person is not moving.

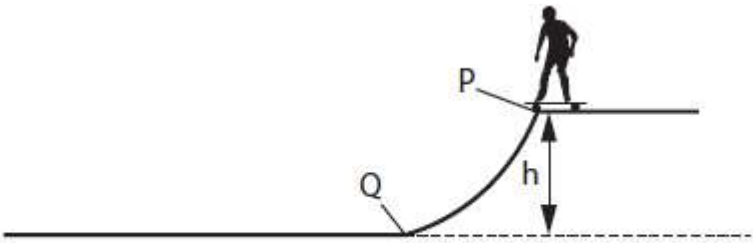

Figure 18

The kinetic energy, KE, of the person at Q is 950 J.

The mass of the person is 35 kg.

Calculate the velocity of the person at Q.

Use the equation

$$v^2 = \frac{2 \times KE}{m}$$

(3)

velocity = ... m/s

(Total for question = 3 marks)

Q7.

Some technicians are investigating factors affecting the braking distance of a bicycle.
They start by investigating how the speed of a bicycle affects the braking distance.

Figure 8 is a graph of their results.

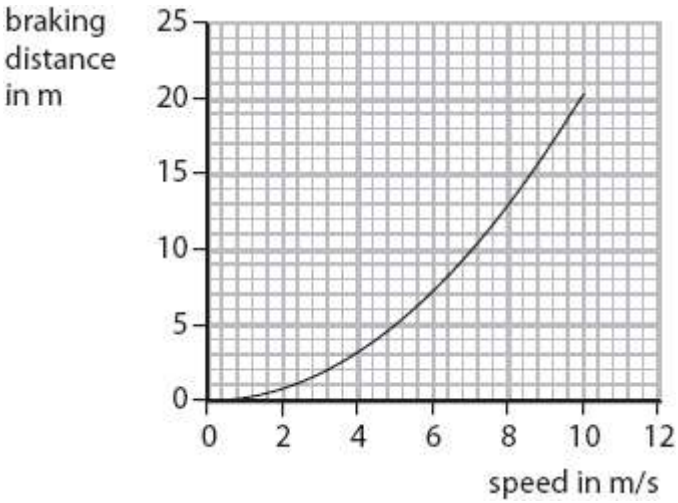

Figure 8

(i) When the bicycle is travelling at a speed of 15 m/s, the kinetic energy (KE) of a cyclist and the bicycle is 9300 J.

Calculate the mass of the cyclist and the bicycle.

Use the equation $m = \frac{2 \times KE}{v^2}$

(3)

mass = ... kg

(ii) Describe the energy transfer that takes place as this cyclist brakes.

(2)

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.....

(Total for question = 5 marks)

Q8.

A model train has a mass of 8.0 kg.
It travels at a speed of 1.5 m/s.

Calculate the kinetic energy of the model train.

Use the equation

$$\text{kinetic energy} = \tfrac{1}{2} \times \text{mass} \times (\text{speed})^2$$

(3)

kinetic energy = ... J

(Total for question = 3 marks)

Q9.

A cyclist of mass 70 kg travels at a constant velocity of 8 m/s.
Calculate the kinetic energy of the cyclist.

(3)

kinetic energy of the cyclist = ... J

(Total for question = 3 marks)

Q10.

Figure 17 shows a football kicked against a wall.

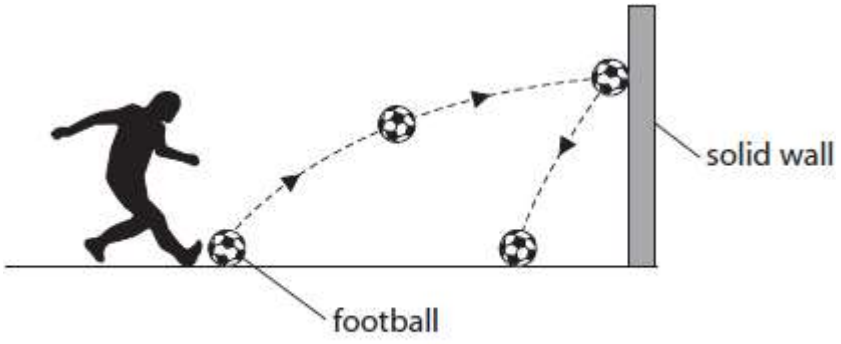

Figure 17

The football has a mass of 0.42 kg.

Calculate the height at which the ball hits the wall.

(3)

Gravitational field strength = 10 N / kg

Use the equation

$$\Delta GPE = m \times g \times \Delta h$$

height = ... m

(ii) Calculate the kinetic energy of the football when it is moving at a velocity of 12 m / s.

(2)

Use the equation

$$KE = \frac{1}{2} \times m \times v^2$$

kinetic energy = ... J

(iii) Describe the energy transfers that happen when the ball hits the wall.

(2)

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(Total for question = 7 marks)

Mark Scheme

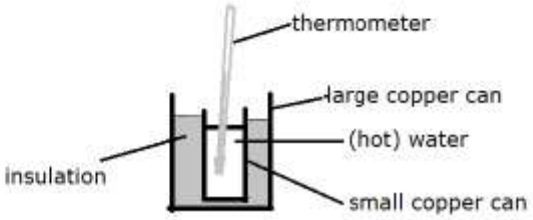
Q1.

Question number	Answer	Additional guidance	Mark
	A description to include: mention relevant energy store such as GPE or chemical (1) 'correct' transfer in context (1)	 allow KE or mechanical or thermal or heat chemical to (G)PE or chemical to KE (in lifting) allow misread GPE to KE/thermal on slope Allow KE to GPE in lifting	**(2)**

Q2.

Q3.

Question Number	Answer	Mark
	B. when there are energy transfers, the total energy does not change ***A*** *is not correct because the total energy does not reduce* ***C*** *is not correct because the total energy does not increase* ***D*** *is not correct because there must be no net change in the total energy*	**(1)** **AO1**

Question Number	Answer	Additional guidance	Mark
(i)	A diagram showing: apparatus labelled to include **three** from • thermometer • water • insulator / sand / sawdust/ material • (copper) can (1) thermometer in the **water** (1) arrangement for water and insulator in and between copper cans (e.g. as in diagram below) (1)  	independent of arrangement ignore kettle and stop clock accept reverse positions for water and insulator	**(3)** **AO2**

Q4.

Question number	Answer	Additional guidance	Mark
(i)	substitution Time = 37/ 25 (1) Evaluation (1) = 1.5 (s)	Allow 1.48 (s) full marks will be awarded for correct numerical answer without working	(2)

Question number	Answer	Additional guidance	Mark
(ii)	substitution K.E. = 0.5 x 1300 x 20^2 (1) evaluation (1) = 260,000 J	260 kJ full marks will be awarded for correct numerical answer without working	(2)

Q5.

Question number	Answer	Additional guidance	Mark
(i)	Substitution (1) PE = 7.26 x 10 x 1.3 Answer (1) 94.4 (J)		(2)

Question number	Answer	Additional guidance	Mark
(ii)	An explanation that combines identification - application of knowledge (1 mark) and reasoning/justification - application of understanding (1 mark): • (energy stored changes between) gravitational potential energy, kinetic energy (1) • Potential energy increases as it rises / decreases as it falls. (1)	Ignore reference to energy change after hitting the ground (eg sound / heat)	(2)

Q6.

Question number	Answer	Additional guidance	Mark
	substitution (1) $(v^2) = \frac{950 \times 2}{35}$ evaluation of v^2 (1) 54(.29) evaluation of v (1) (v =) 7.4 (m/s)	accept values that round to 7.3(m/s) or 7.4(m/s) accept answer of 7 (one sig. fig.) award 2 marks for an answer that rounds to 54 (m/s) if no other mark scored allow 1 mark for an answer that rounds to 0.23 (m/s) (use of mass in g) award full marks for correct answer without working	**3** **AO2.1**

Q7.

Question number	Answer	Additional guidance	Marks
(i)	substitution (1) $(m=)\frac{2\times9300}{15^2}$ 225 for v^2 (1) evaluation (1) 83 (kg)	$(m=)\frac{18600}{15^2}$ 82.7 / 82.6 / 82.67 / 82 (kg) etc. award full marks for correct numerical answer without working	**(3)**

Question number	Answer	Additional guidance	Marks
(ii)	an answer that combines the following points of understanding to provide a logical description: • (as) <u>kinetic energy</u> (store) decreases (1) • (so) thermal energy (store) increases (dissipating to the surroundings) (1)	<u>KE</u> at beginning idea kinetic energy becomes heat (energy) – 2 marks ignore sound for this mark	**(2)**

Q8.

Question Number	Answer	Additional guidance	Mark
	substitution (1) ½ x 8 x 1.5(^{2}) calculation of v^2 (1) 2.25 evaluation (1) 9(.0) (J)	9000 (J) scores 2 marks 6(.0)(J) scores 2 marks 6000 (J) scores 1 mark award full marks for the correct answer without working	**(3)**

Q9.

	Answer	Additional guidance	Mark
	select correct equation (1) $KE = ½ \times m \times v^2$ substitution (1) (KE =) ½ × 70 × 8(.0)[(2)] evaluation (1) (KE =) 2200 (J)	 ignore attempts to convert kg to g for this MP only allow numbers that round to 2200 e.g. 2240 280 or 35 x 8 seen scores 2 marks award full marks for the correct answer without working.	**(3)** **AO2**

Q10.

Question number	Answer	Additional guidance	Mark
(i)	substitution (1) 11 = 0.42 x 10 x Δh rearrangement (1) (Δh =) $\frac{11}{0.42 \times 10}$ evaluation (1) 2.6 (m)	accept substitution and rearrangement in either order (Δh =) $\frac{\Delta GPE}{m \times g}$ accept any value which rounds to 2.6 (m) award 2 marks for 2.6 to any other power of 10 allow 1 mark for 0.38 allow 1 mark for 46(.2) award full marks for the correct answer with no working give full credit for use of g=9.8 or 9.81 N/kg (gives 2.7 (m))	(3) AO2

Question number	Answer	Additional guidance	Mark
(ii)	substitution(1) (KE=)$\frac{1}{2}$ x 0.42 x 12^2 evaluation (1) 30 (J)	allow 30.2(4) (J) award 1 mark for 30 240 (J) award 1 mark for 2.52 (J) award 1 mark for 60.5 (J) award full marks for the correct answer with no working	(2) AO2

Question number	Answer	Additional guidance	Mark
(iii)	A description including: KE/kinetic (energy store) (1) (transfers to) and **one** of: elastic (potential energy store) (1) OR thermal (energy of ball/wall/surroundings) (1) OR <u>dissipates</u> (to surroundings) (1)	allow mechanically / mechanical transfer ignore reference to gravitational potential energy allow heat for thermal allow sound in this context ignore reference to the ground	**(2)** **AO2**